



Overcoming Multidrug Resistance in *E. coli* and *Salmonella* Isolates from Nile Tilapia: Synergistic Effects of Novel Antibiotic Combinations

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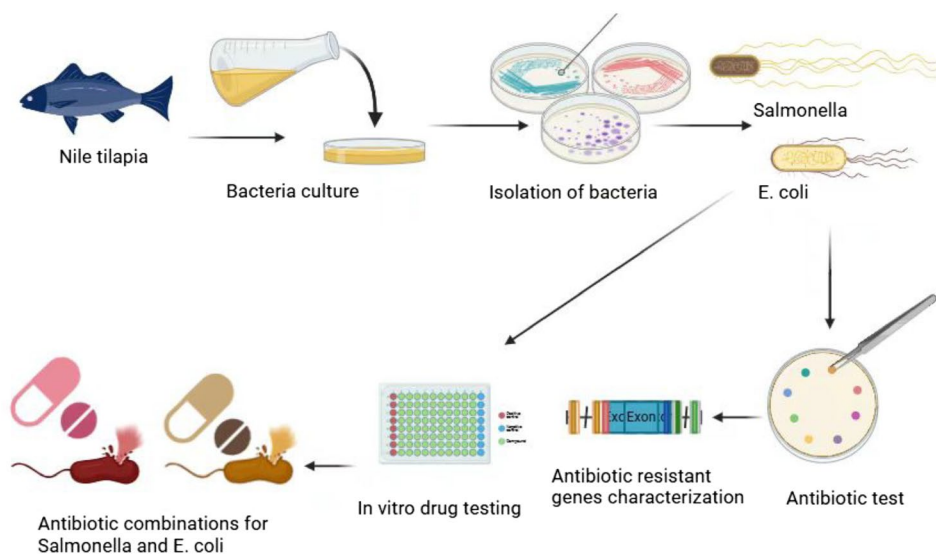
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Abstract

Escherichia coli and *Salmonella* are significant foodborne zoonotic pathogens, causing serious human illness. The rising global prevalence of antimicrobial resistance (AMR) in these species exacerbates their public health risk, complicating the treatment of bacterial infection. This study investigates its prevalence, resistant genes, and treatment strategy against antibiotic-resistant bacteria, focusing on *E. coli* and *Salmonella* isolates from Nile tilapia. Prevalence of *E. coli* and *Salmonella* was found to be 32 and 22% respectively. Antibiotic susceptibility testing revealed resistance to five antibiotics in *E. coli* and four in *Salmonella*. Physiochemical properties of antibiotic resistance genes (ABRGs) indicated that the TetB gene has the highest aliphatic index in both bacteria, suggesting greater stability. All Bla proteins were hydrophobic as indicated by negative GRAVY values, which may contribute to antibiotic efflux or modification of antibiotic targets. Motif analysis identified functional domains, and cellular localization prediction showed that TetA and TetB genes are primarily expressed in the cell membrane. To combat this resistance, a checkerboard method was used to explore novel antibiotic combinations. For *E. coli*, one synergistic and two additive combinations were identified, while for *Salmonella*, two synergistic and one additive combination were effective. These results highlight the importance of regularly evaluating antibiotic combinations to combat resistance and preserve antibiotic efficacy.

Graphical Abstract



Laiba Shafique and Muhammad Farhan Khan contributed equally to this work.

Extended author information available on the last page of the article

Keywords *E. coli* · *Salmonella* · Antibiotic resistance · Minimal inhibitory concentration · Synergism · Additive effect · Checkerboard method

Introduction

Multidrug resistance (MDR), a threat to public health worldwide [1], as this phenomenon significantly limits the effectiveness of conventional treatments, resulting in prolonged infections, increasing mortality rates, and high healthcare costs with rising incidences of MDR, several recent investigations have reported the emergence of MDR bacterial pathogens from different origins [2, 3]. The rapid emergence of MDR foodborne pathogens poses a critical threat to global food safety [4]. A serious concern associated with foodborne illnesses is the rise of antimicrobial resistance (AMR), which is due to the overuse of antibiotics, especially in food production, has accelerated the emergence of MDR bacteria [3, 5].

The rapid expansion of global aquaculture has led to a peak in Nile tilapia production, making it vital for global food security [6]. Tilapia is a valuable protein source, but also functions as a vector for foodborne pathogens [7]. This diversity in production methods surpasses that of any other farmed fish. In addition, recirculating systems have facilitated Nile tilapia farming in urban areas [8]. It has been reported that tilapia could carry various contamination with different foodborne pathogens, like *Plesiomonas shigelloides*, diarrheagenic *Escherichia coli*, *Vibrio parahaemolyticus*, *Salmonella Weltevreden*, *Salmonella enterica*, *Shigella*, *Staphylococcus aureus*, *Campylobacter jejuni*, *Clostridium botulinum*, and *Listeria monocytogenes* [9].

Irresponsible use of antibiotics in Nile tilapia production is a prevalent practice to mitigate bacterial infections and enhance growth [10]. However, this practice raises serious questions regarding AMR along with other contaminations and its potential harmful effects on aquaculture, suggesting effective strategies on fish growth and health [11–13]. Antibiotics such as doxycycline have demonstrated therapeutic efficiency against *Aeromonas hydrophila*, with survival rate notably increasing at higher doses [14]. Likewise, oxytetracycline is associated with alterations in the gut microbiome, potentially increasing the prevalence of resistance genes, which could pose long-term health risks to tilapia [15].

Gram-negative bacteria such as *Escherichia coli* and *Salmonella* can cause a wide range of harmful and potentially life-threatening infections in humans, including gastrointestinal diseases, respiratory infections, urinary tract infections, meningitis, sepsis, and sexually transmitted infections, which can be linked to foodborne illness outbreaks [16–18]. These bacteria are often associated with antibiotic

resistance (ABR), causing difficulties while treating bacterial infections [19]. A substantial proportion of *E. coli* isolates from Nile tilapia are resistant to various antibiotics, with 53.8% showing MDR [20]. Another study revealed the prevalence of ABR in *E. coli* and *Salmonella* from Nile tilapia is 35.29% and 26.47%, respectively, with alarming levels of MDR [21]. Although ABR in aquaculture is well-documented, limited studies have explored the long-term ecological consequences of antibiotic overuse in Nile tilapia farming, as well as alternative strategies to mitigate AMR [22]. Therapies in combination with antibiotics, such as florfenicol and enrofloxacin showed effective results against inflammation as well as an improved immune response in Nile tilapia [23]. Furthermore, the in vivo study showed that alpha-linolenic acid (ALA) combined with florfenicol (FFC) or tetracycline (TET) minimized the MIC of both antibiotics against MDR *Salmonella Typhimurium* [24].

AMR is an evolving threat that challenges the development of new antibacterial agents, creating a significant gap in addressing existing treatment techniques, with global data limitations highlighting the urgent need for ongoing research and assessment to better understand the extent of ABR in various contexts [25].

Materials and Methods

Sample Collection

265 Samples of Nile tilapia were collected from 10 different Qinzhou aquatic product markets and seafood supermarkets, and each fish sample was individually packed in a clean polyethylene bag. These samples were preserved in an ice box maintained at 4 °C and shifted to the Guangxi State Key Laboratory of Beibu Gulf University.

Isolation and Identification of Bacteria

Under sterile conditions, after wiping the surface of the fish body with 70% alcohol for disinfection, 0.2–0.3 g homogenized samples were taken from the flesh and gills. A total of 265 samples of Nile tilapia (muscle, tissue) were further processed for the isolation and identification of *E. coli* and *Salmonella*. The samples were incubated overnight at 37 °C in sterile nutrient broth. Incubated samples were homogenized using a stirrer, and sterile swabs were immersed in the homogenized material [26]. Swabs were streaked onto

blood agar and incubated at 37 °C for 24 h. Extracted colonies were subsequently streaked on MacConkey agar for *E. coli* and *Salmonella*–*Shigella* agar for *Salmonella*, incubated at 37 °C for 24 h, respectively. Pure colonies were processed for identification.

A series of biochemical tests (gram staining and IMViC test) was conducted to identify *E. coli* and *Salmonella*, by following the standard protocols outlined in Bergey's Manual of Determinative Bacteriology [27].

Antibiogram

To assess the susceptibility of *E. coli* and *Salmonella*, a total of 14 antibiotics were tested (amoxicillin, enrofloxacin, tetracycline, sulfamethoxazole, ampicillin, amikacin, kanamycin, ciprofloxacin, chloramphenicol, cefoxitin, erythromycin, doxycycline, gentamicin, and vancomycin). The Kirby–Bauer disc diffusion technique was used under the guidelines of the Clinical and Laboratory Standards Institute, CLSI [28]. Briefly, fresh *E. coli* culture adjusted to a concentration of 1–1: 5×10^8 CFU/ml and spread onto sterile Mueller–Hinton agar. Antibiotic discs were aseptically placed at equidistant points on the agar surface. After incubation at 37 °C for 20–24 h, the zones of inhibition were measured and compared to the standards provided by CLSI [28].

Molecular Characterization in Antibiotic Resistance Genes

The NCBI database was used to retrieve sequences of the resistance genes in both bacteria. Physicochemical properties, multiple sequence alignment motifs, gene structure, and cellular distribution were analyzed.

Physicochemical Properties

The ProtParam tool on ExPASy (<https://web.expasy.org/protparam/>) was utilized to compute the physical and chemical properties, including molecular weight (M_w), isoelectric point (pI), instability index (I_i), aliphatic index (A_i), and grand average of hydropathy (GRAVY). In addition, chromosomal numbers were calculated by referencing genomic sequences available in the NCBI database [29].

Multiple Sequence Alignment

Domain sequences of selected genes were aligned using the Multiple Align Display (<https://www.bioinformatics.org/sms/multialign.html>) to identify variant sequences. The following method allows for the identification of functionally significant regions within proteins and further facilitates the inference of evolutionary relationships [30].

Motif Analysis

MEME suite was employed to identify and analyze 10 conserved MEME motifs of the selected proteins. For this analysis, the selected protein file was prepared and subsequently loaded into the MEME platform. The resulting XML file was then utilized for further in-depth examination of the identified motif patterns [31].

Secondary Structure

The amino acid sequences of the selected genes from the represented species were uploaded to the Phyre2 online server for the prediction of their 3D structures. This platform enabled the generation of detailed three-dimensional models, along with the identification of secondary structure elements like alpha-helices and beta-sheets. Advances in algorithms of Phyre2 were utilized to evaluate the structural characteristics based on sequence homology. These predictions offered an understanding of the structural and functional properties of the protein [32]. Cellular localization of selected genes was identified by Cello-life (<http://cello.life.nctu.edu.tw/>) and Wolf Sort (<https://wolfsort.hgc.jp/>), the subcellular localization [33].

In Vitro Therapeutics for Testing Antibiotics

The broth microdilution method was utilized to calculate the minimal inhibitory concentration (MIC) of different antibiotic preparations. The checkerboard method was used to evaluate the combination therapeutics [34].

Minimal Inhibitory Concentration

To evaluate this, 96-well titration plates were filled with Mueller–Hinton broth, and the preparations were serially diluted using a twofold dilution method. Fresh cultures of *E. coli* and *Salmonella* were added to each well with adjustment of 10^5 CFU/ml, excluding the negative control. The negative control contained only broth, while the positive control contained both culture and broth. Optical density (OD) readings were computed at a wavelength of 570 nm using an ELISA reader at both 0 h and 24 h. The net OD value was computed by comparing the 24-h readings with the 0-h baseline and then used to evaluate bacterial growth inhibition at different concentrations. MIC was defined as the lowest concentration of preparation that effectively inhibited bacterial growth [35].

Synergy Testing

Drug combinations against isolates of *E. coli* and *Salmonella* (1 – 1.5×10^5) were evaluated for synergy using the

checkerboard method. The OD value of the 96-well plates was computed at 570 nm both before and after a 24-h incubation at 37 °C. The inhibitory concentration indices were measured using the specified formula. To minimize errors, the experiment was conducted in triplicate [35].

Fractional Inhibitory Concentration Index (FICI) = Fractional Inhibitory Concentration (FIC) of product A + FIC of product B,

FIC of product A = MIC of product A in combination with product B/MIC of product A alone,

FIC of product B = MIC of product B in combination with product A/MIC of product B alone,

FICI ≤ 0.5 was considered as synergistic; > 0.5 but ≤ 1 as an additive; > 1 but < 4 as indifferent; and > 4 as antagonistic.

Statistical Analysis

Descriptive statistics and one-way ANOVA were applied, and statistical analysis was performed using SPSS software (version 20) at a 5% significance level. For physicochemical properties, multiple sequence alignment motifs, gene structure, and cellular distribution bioinformatics tools (ExPASy, MSA, Tb tool, Phyre) were applied to data sets.

Results

Prevalence of *E. coli* and *Salmonella*

The prevalence rates for *E. coli* and *Salmonella* were identified to be 32% and 22%, respectively. Statistical analysis showed that the prevalence of both pathogens was statistically significant (p value = 0.009) shown in Table 1.

Antibiogram of *E. coli* and *Salmonella* Isolates

The discs diffusion assay revealed that *E. coli* isolates exhibited high-level resistance to sulfamethoxazole (73.33%); amoxicillin (60%); chloramphenicol, ampicillin, and tetracycline (53.33%); ciprofloxacin (40%); cefoxitin (33.33%); enrofloxacin (26.67%); erythromycin (26.66%); gentamicin, amikacin, and kanamycin (20%); doxycycline (13.33%); and, vancomycin (0%) (Table 2). The overall resistance profile of the *E. coli* isolates was as stated: sulfamethoxazole $>$ amoxicillin $>$ chloramphenicol, ampicillin and tetracycline $>$ ciprofloxacin $>$ cefoxitin $>$ enrofloxacin $>$ erythromycin $>$ gentamicin, amikacin, and kanamycin $>$ doxycycline $>$ vancomycin.

Similarly, *Salmonella* isolates exhibited high resistance to ampicillin (73.33%); sulfamethoxazole (60%); erythromycin (53.33%); amoxicillin (46.67%); chloramphenicol, kanamycin, cefoxitin and tetracycline (40%); amikacin (26.67%); erythromycin (26.66%); gentamicin, ciprofloxacin and enrofloxacin (20%); doxycycline (13.33%); and, vancomycin (6.67%) (Table 3). The overall resistance order of the *Salmonella* isolates was as follows: ampicillin $>$ sulfamethoxazole $>$ erythromycin $>$ amoxicillin $>$ chloramphenicol, kanamycin, cefoxitin, and tetracycline $>$ amikacin $>$ erythromycin $>$ gentamicin, ciprofloxacin and enrofloxacin $>$ doxycycline $>$ vancomycin.

Molecular Characterization of Antibiotic Resistance Genes

To analyze the ABR in *E. coli* and *Salmonella*. The data on selected resistant genes were exported from the NCBI database, and comparative analysis was performed between both species on the genomic level (Supplementary Table 1).

Table 1 Prevalence of *E. coli* and *Salmonella* isolated from Nile tilapia

No	Total tilapia samples	Prevalence <i>E. coli</i>	Negative	%Age	p Value	Prevalence <i>Salmonella</i>	Negative	%Age	p Value
Dongfong	35	9	26	25.71	0.072	8	27	22.86	0.922
Dalu Street	25	7	18	28	0.009	4	21	16	0.009
Wuyue Mart	20	7	13	35		4	16	20	
Walmart	20	3	17	15		4	16	20	
Xinxing Market	30	6	24	20		7	23	23.33	
Chuanda Market	20	4	16	20		5	15	25	
Chengzhongnan Market	25	11	14	44		6	19	24	
Qili Xingxi Street	35	17	18	48.57		5	30	14.29	
Chengdong Market	30	14	16	46.67		8	22	26.67	
Heshun Market	25	8	17	32		8	17	32	
Total	265	86	179	32.45		59	206	22.26	

Table 2 Antibiotic susceptibility profile against *E. coli*

Antibiotics	Resistance	Intermediate	Sensitive	Resistance	Intermediate	Sensitive	<i>p</i> Value
Amoxicillin	9	4	2	60	26.67	13.33	0.505
Enrofloxacin	4	8	3	26.67	53.33	20	
Tetracycline	8	1	6	53.33	6.67	40	
Gentamicin	3	7	5	20	46.67	33.33	
Doxycycline	2	3	10	13.33	20	66.67	
Cefoxitin	5	4	6	33.33	26.67	40	
Vancomycin	0	1	14	0	6.67	93.33	
Ciprofloxacin	6	4	5	40	26.67	33.33	
Chloramphenicol	8	2	5	53.33	13.33	33.33	
Amikacin	3	1	11	20	6.67	73.33	
Kanamycin	3	3	9	20	20	60	
Erythromycin	4	6	5	26.66	40	33.33	
Ampicillin	8	4	3	53.33	26.67	20	
Sulfamethoxazole-Trimethoprim	11	1	3	73.33	6.67	20	

Table 3 Antibiotic susceptibility profile against *Salmonella*

Antibiotics	Resistance	Intermediate	Sensitive	Resistance	Intermediate	Sensitive	<i>p</i> Value
Amoxicillin	7	3	5	46.67	20	33.33	0.191
Enrofloxacin	3	6	6	20	40	40	
Tetracycline	6	4	5	40	26.67	33.33	
Gentamicin	3	4	8	20	26.67	53.33	
Doxycycline	2	3	10	13.33	20	66.67	
Cefoxitin	6	4	5	40	26.67	33.33	
Vancomycin	1	1	13	6.67	6.67	86.67	
Ciprofloxacin	3	7	5	20	46.67	33.33	
Chloramphenicol	6	3	6	40	20	40	
Amikacin	4	6	5	26.67	40	33.33	
Kanamycin	6	6	3	40	40	20	
Erythromycin	8	4	3	53.33	26.67	20	
Ampicillin	11	1	3	73.33	6.67	20	
Sulfamethoxazole-Trimethoprim	10	2	3	66.67	13.33	20	

Physicochemical Properties

In this study, we investigated the Tet, Sul, and Bla gene families in the genomes of both *E. coli* and *Salmonella*. The physical and chemical properties of these genes were analyzed, including M_w , I_1 , A_1 , pI, and GRAVY as presented in Table 4a–c. For both *E. coli* and *Salmonella*, Tet proteins exhibited lengths ranging from 138 to 639 amino acid residues. TetA and TetB are more hydrophobic with GRAVY values greater than 0, while TetC, TetD, and TetM have negative GRAVY values, indicating that they are hydrophilic. TetB has the highest A_1 in both *E. coli* (122.64) and *Salmonella* (123.62), demonstrating a higher

stability at elevated temperatures. The Sul gene family in *E. coli* and *Salmonella* showed the following outcomes: Sul2 in *Salmonella* has the highest M_w (31,795.52) as compared to other Sul genes in both represented species, whereas Sul3 is the smallest with M_w 28,898.40. Sul2 in both species demonstrates a higher pI with *Salmonella* having a value of 8.63, which is more alkaline compared to other Sul proteins in both species, while Sul1 and Sul3 have values around 5.5–5.8, suggesting that they are more neutral to slightly acidic. Sul1 shows a higher I_1 in both *E. coli* (49.74) and *Salmonella* (48.61), indicating less stability as compared to Sul2 and Sul3. Sul2 exhibits higher A_1 in both species, depicting higher thermal stability. Sul2

Table 4 Physicochemical properties of ARGs in *E. coli* and *Salmonella*

Species	Gene	M_w	AA	pI	I_I	A_I	GRAVY
(a) Tet family							
<i>Escherichia coli</i>	TetA	42,240.93	399	9.41	39.77	113.13	0.699
	TetB	43,356.21	401	5.95	35.66	122.64	0.812
	TetC	25,770.89	222	8.84	30.20	103.69	-0.211
	TetD	16,774.42	138	9.92	59.38	89.06	-0.572
	TetM	72,589.30	639	5.71	38.08	96.98	-0.237
<i>Salmonella</i>	TetA	45,157.27	424	9.94	42.19	111.96	0.600
	TetB	43,237.18	401	6.97	35.75	123.62	0.845
	TetC	22,746.29	197	8.43	28.15	103.50	-0.257
	TetD	16,774.42	138	9.92	59.38	89.06	-0.572
	TetM	72,782.62	639	5.87	36.85	98.67	-0.220
(b) Sul family							
<i>Escherichia coli</i>	Sul1	30,126.25	279	5.76	49.74	96.92	-0.068
	Sul2	28,469.54	271	5.95	38.10	99.96	0.125
	Sul3	28,898.40	263	5.53	31.53	97.87	-0.005
<i>Salmonella</i>	Sul1	30,098.20	279	5.76	48.61	97.96	-0.071
	Sul2	31,795.52	298	8.62	37.57	98.72	0.128
	Sul3	28,898.40	263	5.53	31.53	97.87	-0.005
(c) Bla family							
<i>Escherichia coli</i>	Bla_TEM	31,515.20	286	5.69	40.74	93.53	-0.109
	Bla_SHV	27,090.07	248	7.97	39.28	95.28	-0.160
	Bla_CTV	20,945.85	197	9.04	21.89	85.28	-0.186
	Bla_Ampc	23,021.26	209	8.98	31.96	82.20	-0.252
<i>Salmonella</i>	Bla_TEM	14,649.72	134	5.68	32.38	90.45	-0.301
	Bla_SHV	14,513.59	135	5.62	35.46	90.44	-0.034
	Bla_CTV	31,148.53	291	8.90	27.02	90.62	-0.179
	Bla_Ampc	42,003.33	381	9.15	32.86	87.11	-0.167

in both species is the most hydrophobic, with GRAVY of 0.125 (*E. coli*) and 0.128 (*Salmonella*) compared to Sul1 and Sul3 with negative or near-zero GRAVY values, which indicates that Sul2 has characteristics of membrane proteins. In the Bla gene family, *E. coli* Bla_TEM and Bla_SHV have larger M_w compared to their *Salmonella* counterparts, with values around 31,000–32,000 Da. Conversely, Bla_CTV and Bla_Ampc show larger M_w in *Salmonella* (31,000 and 42,000 Da) as compared to *E. coli* (21,000 and 23,000 Da). *Salmonella* generally shows a similar or slightly lower pI than *E. coli*, except for Bla_CTV and Bla_Ampc, which have higher pI values in *Salmonella*. *E. coli* Bla_TEM has a higher I_I as compared to other proteins in both species, indicating that it may be more prone to degradation. The A_I values are relatively close across both species, with Bla_SHV demonstrating the highest in *E. coli* (95.28) and Bla_SHV in *Salmonella* (90.44), suggesting a potentially high stability related to hydrophobic interactions. All Bla proteins are hydrophobic as indicated by negative GRAVY values. The Bla_TEM

and Bla_SHV proteins in both represented species show more negative GRAVY, indicating higher hydrophilicity compared to other Bla proteins.

Multiple Sequence Alignment

To identify closely related proteins, we used multiple sequence alignment (MSA) for the identification of evolutionary relationships and functional patterns. To analyze the sequence parameters of the selected genes in *E. coli* and *Salmonella*, outcomes from MSA showed that the prominent red color around conserved amino acid regions depicts the importance of these regions functionally and structurally. Alignment gaps represent green color, whereas yellow colors represent nonconserved amino acids. Based on the presented data, many alignments demonstrated the commonalities, differences, and indels between the represented species (Supplementary Fig. 1).



Fig. 1 **A** Motif prediction of the tetracycline-resistant gene family in *E. coli* and *Salmonella*. **B** Motif prediction of the sulfonamide resistant gene family in *E. coli* and *Salmonella*. **C** Motif prediction of the Bla resistant gene family in *E. coli* and *Salmonella*

Motif Analysis

The Tet gene family in *E. coli* and *Salmonella* exhibits significant structural and functional similarities with conserved motifs reflecting shared evolutionary pressure and critical

roles in ABR mechanisms (Fig. 1A). In both species, TetA and TetB display the most complex motif architectures, consistently sharing conserved motifs (1, 2, 3, 7, and 9), indicating their essential and evolutionary conserved functions. TetC and TetD demonstrate simpler architectures with

TetD consistently containing motifs 2 and 5, while TetE remains the shortest and least complex protein, composed solely of motifs 5 and 10, which suggests a specialized or truncated function linked to environmental or physiological requirements. On the other side, subtle differences in motif arrangement or utilization between the two represented species may reflect species-specific adaptations to distinct ecological niches and selective pressure. Although conserved motifs indicate overlapping functions across *E. coli* and *Salmonella*, unique adaptations could finetune protein functions to better respond to environmental challenges like exposure to tetracycline antibiotics. These differences underline the flexibility of the Tet gene family, emphasizing its structural and functional diversity. Overall, the conserved motifs highlight shared essential roles, while variations in motif utilization reflect evolutionary strategies employed by *E. coli* and *Salmonella* to optimize survival and fitness under different ecological and physiological conditions. This analysis underscores the evolutionary versatility and biological significance of the Tet gene family in these two bacterial species. The Sul gene family in both represented species exhibits a highly conserved motif composition and structural organization, underscoring their shared evolutionary origins and critical roles in sulfonamide resistance (Fig. 1B). Conserved motifs, such as 1, 3, 5, 6, 7, 9, and 10 are consistently present across both species, reflecting their essential functional roles in ABR mechanisms. Although the overall motif architecture is remarkably similar. Elusive differences in motif arrangement, specifically in Sul2 of *Salmonella*, indicate species-specific regulatory adaptations. The motif analysis of the Bla gene family in *E. coli* and *Salmonella* highlights the conserved features and species-specific adaptations (Fig. 1C). In both bacterial species, Bla_TEM, Bla_SHV, and Bla_CTX share conserved motifs (1, 3, 4, and 5), underscoring their critical roles in Bla resistance. However, *E. coli* shows higher motif complexity in Bla_TEM and Bla_SHV with additional motifs such as 7 and 9, potentially enhancing resistance capabilities. In contrast, *Salmonella* exhibits simpler motif arrangements in Bla_TEM, while Bla_SHV and Bla_CTX feature unique motifs such as 6 and 10, indicating adaptations to distinct selective pressures. Bla_Ampc remains the least complex and shortest protein in both representative species, retaining minimal motifs such as 2.3 and 5, demonstrating a specialized but conserved function. These differences emphasize the evolutionary flexibility of varying ecological and antibiotic challenges in *E. coli* and *Salmonella*.

Secondary Structure

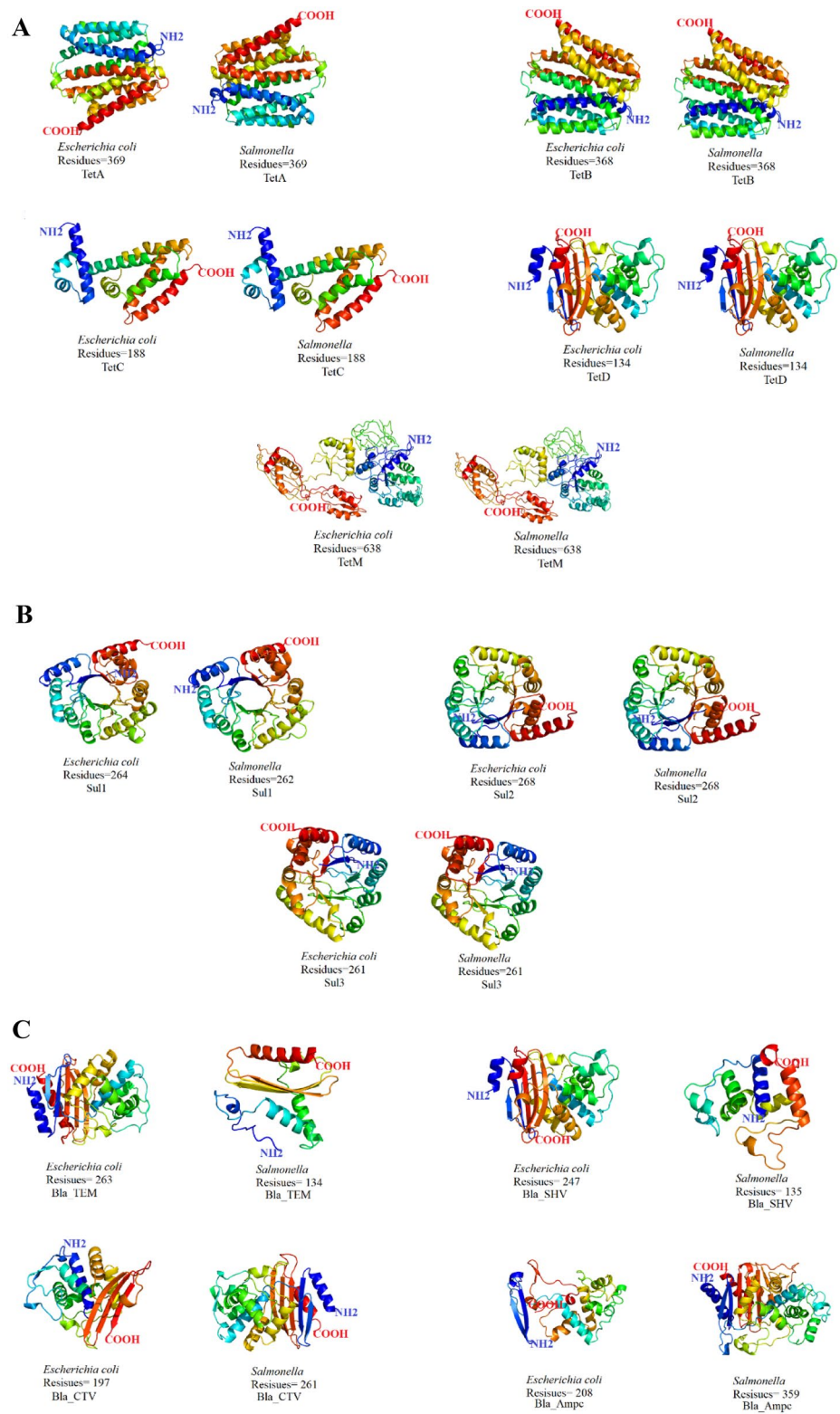
The secondary structure of the three gene families, Tet, Sul, and Bla, in *E. coli* and *Salmonella* was analyzed. The Tet proteins show significant structural conservation

between *E. coli* and *Salmonella*, with TetA, TetB, TetC, and TetD playing key roles as these proteins are involved in mechanisms of ABR and reveal similar functional features in both bacteria (Fig. 2A). Moreover, the Sul proteins exhibit high structural and residue conservation between both represented species, with Sul1 differing slightly in residue count (264 in *E. coli* and 262 in *Salmonella*) as shown in Fig. 2B. In addition, the Bla gene family displays greater differences in residue count and structure between *E. coli* and *Salmonella*, indicating functional divergence, with Bla_TEM and Bla_SHV showing notable variations (Fig. 2C). The aforementioned three gene families in *E. coli* and *Salmonella* reveal different levels of structural conservation and divergence, and reflect their distinct functional roles and adaptations. The Tet and Sul gene families exhibit high structural and residue conservation between two bacterial species which suggests a shared evolutionary pathway and consistent functions like antibiotic efflux (Tet) and sulfonamide resistance (Sul). Conversely, the Bla gene family shows remarkable differences in residue count and structural complexity, with some genes being truncated, such as Bla_TEM and Bla_SHV, or expanded, like Bla_CTX and Bla_Ampc in *Salmonella*, indicating more pronounced functional divergence. These variations across the three gene families highlight the differing evolutionary pressure and resistance mechanisms employed by *E. coli* and *Salmonella*.

Cellular Localization Prediction

The Tet, Sul, and Bla gene families in *E. coli* and *Salmonella* exhibit distinct localization patterns and expression levels across different cellular compartments. The Bla gene family exhibits high variability between both bacterial species, with *Salmonella* displaying a higher expression of Bla_Ampc in the extracellular and peroxisomal sections, whereas *E. coli* has a more balanced distribution across the cytoplasm, mitochondria, and extracellular regions for Bla_TEM and Bla_CTX. The Sul gene family demonstrates similar distribution in both species, with Sul1 being highly localized in the cytoplasm and Sul2 revealing strong extracellular presence in both species. The Tet gene family exhibits remarkable conservation between the two bacterial species, specifically for TetA and TetB which show predominant localization in the plasma membrane, whereas TetC, TetD, and TetM are more concentrated in the cytoplasm and nucleus. These patterns indicate that the Sul and Tet families maintain conserved roles across the two species, and the Bla family shows greater species-specific adaptations, likely connected to differences in Bla resistance mechanisms. The heatmap depicts diverse expression patterns of the Tet, Sul, and Bla gene families in both bacterial species, as shown in Fig. 3.

Fig. 2 **A** Secondary structure analysis of the tetracycline gene family in *E. coli* and *Salmonella*. **B** Secondary structure analysis of sulfonamide gene family in *E. coli* and *Salmonella*. **C** Secondary structure analysis of the Bla gene family in *E. coli* and *Salmonella*



Minimum Inhibitory Concentration

The MIC value is important for evaluating the effectiveness of antibiotics. The following antibiotics (amoxicillin,

tetracycline, sulfamethoxazole, ampicillin, and chloramphenicol) were selected for monitoring resistant trends to ensure the antibiotic usage to treat bacterial (*E. coli* and *Salmonella*) infection. The MICs of both bacterial species

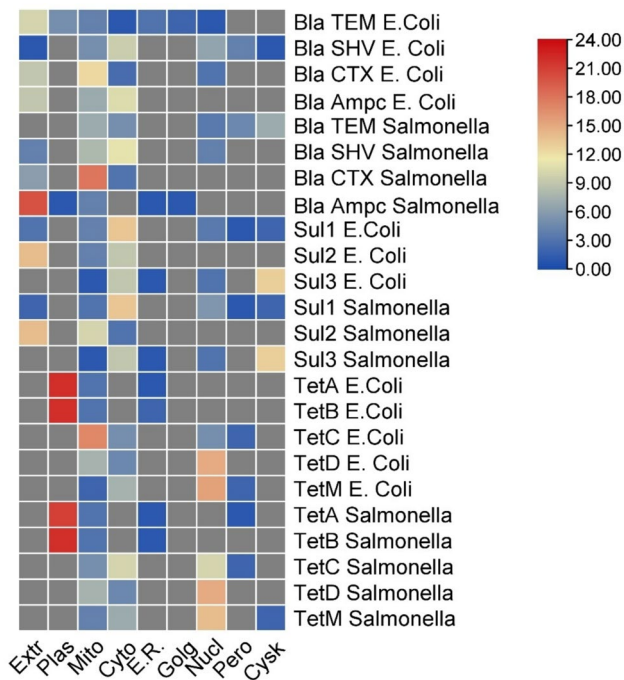


Fig. 3 Distribution of ARGs in *E. coli* and *Salmonella* in the cell and their organelles

exhibit significant differences in their ABR profile (Tables 5, 6). Both bacterial species were tested for the MIC value. The MIC value of amoxicillin against *E. coli* was (4.25 µg/ml), and *Salmonella* showed (18 µg/ml). The MIC value of tetracycline against *E. coli* was 160 µg/ml, and *Salmonella*

showed 22 µg/ml. The MIC value of ampicillin against *E. coli* was 146.5 µg/ml, and *Salmonella* showed 5.38 µg/ml. The MIC value of Sulfamethoxazole against *E. coli* was (474.50 µg/ml), and *Salmonella* showed (320 µg/ml). The MIC value of chloramphenicol was 8.85 µg/ml against *E. coli*. The overall trend for *E. coli* was as follows: sulfamethoxazole > ampicillin > tetracycline > chloramphenicol > amoxicillin; and the overall trend in *Salmonella* was as stated: sulfamethoxazole > tetracycline > amoxicillin > ampicillin. These findings proposed the antimicrobial mechanisms, including cell membrane damage, alteration of membrane permeability, inhibition of nucleic acid and protein synthesis, reduced expression of virulence factors, mitochondrial dysfunction, and prevention of biofilm formation with tested drugs.

Synergy Evolution of Antibiotics

The FICI is a valuable tool for evaluating the interaction between antimicrobial agents, identifying effective drug combinations, and optimizing treatment strategies. It plays a key role in combating AMR, improving clinical outcomes, and guiding research into new therapeutic approaches for resistant infections. The fractional inhibitory indices of antibiotics alone and with combinations revealed distinct interactions. For *E. coli*, synergistic effects were observed which was in the combination of sulfamethoxazole + ampicillin, while antagonistic interactions were observed in the combinations of amoxicillin + tetracycline and tetracycline + sulfamethoxazole. An additive response

Table 5 Minimum inhibitory concentration of commonly used antibiotics against *E. coli*

<i>E. coli</i>	Mean	Standard deviation	95% Confidence interval for mean		<i>p</i> Value
			Lower bound	Upper bound	
Amoxicillin	4.25	1.85	1.31	7.19	0
Tetracycline	160	82.62	28.53	291.47	
Chloramphenicol	8.85	2.42	5.00	12.70	
Sulfamethoxazole	474.50	209.01	141.92	807.08	
Ampicillin	146.50	53.90	60.74	232.26	

Table 6 Minimum inhibitory concentration of commonly used antibiotics against *Salmonella*

<i>Salmonella</i>	Mean	Standard deviation	95% Confidence interval for mean		<i>p</i> Value
			Lower bound	Upper bound	
Amoxicillin	18	10.58	1.16	34.84	0
Tetracycline	22	5.164	13.78	30.22	
Sulfamethoxazole	320	165.25	57.06	582.95	
Ampicillin	5.38	2.057	2.10	8.65	

Table 7 Synergy testing of antibiotics against *E. coli*

<i>E. coli</i>									
Plate no	Combination	Antibiotic alone MIC	Antibiotic in combination	FIC A	2d Antibiotic alone MIC	2d a combination	FIC B	FICI	Remarks
1	amoxi + tetra	4.25	16	3.76	160	256	1.6	5.36	Antagonistic
2	amoxi + chloram	4.25	1.33	0.31	8.85	5.33	0.60	0.92	Additive
3	amoxi + sulfa	4.25	2	0.47	474.5	128	0.27	0.74	Additive
4	amoxi + ampi	4.25	4	0.94	146.5	128	0.87	1.81	Indifferent
5	tetra + chloram	160	128	0.8	8.85	10.67	1.21	2.01	Indifferent
6	tetra + sulfa	160	384	2.4	474.5	768	1.62	4.02	Antagonistic
7	tetra + ampi	160	128	0.8	146.5	256	1.75	2.55	Indifferent
8	chloram + sulfa	8.85	8	0.903955	474.5	384	0.81	1.71	Indifferent
9	chloram + ampi	8.85	5.33	0.60226	146.5	85.33	0.58	1.18	Indifferent
10	sulfa + ampi	474.5	128	0.269758	146.5	32	0.22	0.49	Synergistic*

Amoxi Amoxicillin, *tetra* Tetracycline, *sulfa* Sulfamethoxazole, *ampi* Ampicillin, *chloram* Chloramphenicol

was observed in the combinations of amoxicillin + chloramphenicol and amoxi + sulfamethaxazole. The remaining combination showed an indifferent response as shown in Table 7. On the same pattern, synergistic effects were observed in *Salmonella*, which was in the combination of amoxicillin + sulfamethoxazole and ampicillin + sulfamethoxazole. In contrast, antagonistic interactions were noted for the combination of tetracycline and ampicillin. An additive effect was identified for the combination of amoxicillin and ampicillin. Amoxicillin + tetracycline and tetracycline + sulfamethoxazole exhibited an indifferent response as summarized in Table 8.

Discussion

Antimicrobial agents are essential for minimizing mortality and morbidity related to inflammatory ailments in humans and animals [36]. The overuse of antimicrobials in humans

and animals has promoted antimicrobial-resistant bacteria [37]. Therefore, this study was designed to investigate the prevalence and ABR of commonly used antibiotics against *E. coli* and *Salmonella*. In addition, provides knowledge about ABRGs characterization in both species and also highlights variations between ABRGs. Finally, our study provides a combination therapy to treat antimicrobial infection.

The key findings of our study revealed that the prevalence of *E. coli* and *Salmonella* has been observed at 32% and 22%, respectively. The prevalence of bacteria has become so high nowadays due to contamination. Humans can be infected with AMR bacteria by eating aquatic animals or having direct contact with a contaminated environment. Resistant bacteria identified in this study are considered important estuarine environmental pollutants that can adversely affect food security and public health, and high prevalence of *E. coli* and *Salmonella* has also been observed in previous literature [38]. In our study, sulfamethoxazole showed high resistance in both bacterial isolates, and similar trends were

Table 8 Synergy testing of antibiotics against *Salmonella*

<i>Salmonella</i>									
Plate no	Combination	Antibiotic alone MIC	Antibiotic in combination	FIC A	2d antibiotic alone MIC	2d a combination	FIC B	FICI	Remarks
1	amoxi + tetra	18	48	2.67	22	32	1.45	4.12	Indifferent
2	amoxi + sulfa	18	4	0.22	320	64	0.2	0.42	Synergistic*
3	amoxi + ampi	18	4	0.22	5.38	4	0.74	0.97	Additive
4	tetra + sulfa	22	32	1.45	22	48	2.18	3.64	Indifferent
5	tetra + ampi	22	48	2.18	5.38	28	5.21	7.39	Antagonistic
6	sulfa + ampi	320	64	0.2	5.38	1	0.19	0.39	Synergistic*

Amoxi Amoxicillin, *tetra* Tetracycline, *sulfa* Sulfamethoxazole, *ampi* Ampicillin

observed in the following study [38]. Previous studies exhibited significant resistance to commonly used antibiotics, which is also in line with our study findings [39, 40]. The high prevalence of aquatic pathogens and significant AMR raises a very important question about treatment complications for infections in humans, making it a pressing public health concern. The interconnectedness of aquaculture with human and animal health necessitates a comprehensive approach to mitigate these risks [41]. Aquaculture is widely regarded as one of the most promising sectors within the global food production industry. The transition from semi-intensive to intensive farming techniques, along with the use of antibiotics to manage disease outbreaks, are the key factors for growth in aquaculture production. Antibiotics are usually used at subtherapeutic levels in an aquaculture setting, which enhances the selective pressure to resistant bacteria and activates resistant genes [42]. The following drugs, Tet (tetracycline resistance), Sul (sulfonamide resistance), and Bla (beta-lactam resistance) are used to treat bacterial infection in fishes, the presence of these resistance genes in bacterial populations may compromise the efficiency of commonly used antibiotics, resulting in treatment failure. The physical and chemical properties of genes, such as M_w , pI, GRAVY, etc., play a crucial role in comprehending their functionality and biochemical behavior. These characteristics have an impact on protein structure, stability, and interaction, thereby influencing gene expression and function [43]. In line with the findings of [29], our study showed that Tet proteins showed varied hydrophobicity, whereas Sul and Bla proteins were more stable and hydrophobic. On the other hand, Sul and Bla gene families highlight a remarkable difference in M_w , pI, and instability, with Sul2 and Bla_TEM showing distinct characteristics related to protein stability and membrane interactions. The A_I , representing the proportion of protein volume occupied by aliphatic side chains (alanine, valine, isoleucine, and leucine), can be seen as a positive factor in enhancing protein thermostability. Since the A_I correlated with protein thermostability, a higher A_I value indicates greater protein thermostability [44]. An A_I value higher than 65 indicates high stability [45]. Likewise, in our findings, TetB A_I value is 122.64 (> 65) for *E. coli* and 122.64 (> 65) for *Salmonella*, suggesting greater stability. Owing to the presence of stable resistance genes on plasmids, transposons, or integrons, bacteria may spread via horizontal gene transfer (HGT), a key mechanism by which ABR spread to the bacterial population [46].

According to our study findings, Tet, Sul, and Bla gene families in *E. coli* and *Salmonella* shared conserved motifs that highlight their critical roles in ABR. Species-specific variations in motif composition and arrangement depict evolutionary adaptations to distinct ecological and physiological challenges, illustrating the versatility of selected gene families in optimizing survival and fitness. Similar

studies have been reported that are in agreement with the findings of our study [47]. The secondary structure of resistance genes plays a vital role in maintaining their stability, functionality, and interactions with antibiotics, influencing the mechanism of drug resistance across bacterial species [48]. Our study findings are similar to the previous literature in which Tet and Sul proteins in *E. coli* and *Salmonella* exhibit high structural and residue conservation, indicating shared evolutionary pathways and consistent functions such as antibiotic efflux and Sul resistance [49]. In contrast, the Bla proteins revealed greater divergence and distinct resistance mechanisms between the two bacterial species [50, 51]. These findings highlight the varying evolutionary pressure and adaptation strategies of both bacterial species in response to ABR, emphasizing the complexity and diversity of resistance mechanisms within different gene families [52]. As per the previous literature, our study also reported a similar trend of localization predictions [29], the Tet and Sul proteins showed conserved localization patterns in both bacterial species, with TetA and TetB mainly in the plasma membrane and Sul1 in the cytoplasm, indicating a shared functional role. In contrast, the Bla proteins exhibit more species-specific expression with *Salmonella*, revealing higher Bla_Ampc levels in extracellular and peroxisomal regions, depicting distinct adaptations in Bla resistance [53, 54]. These differential localization and expression patterns across the three gene families indicate that both bacterial species employ distinct strategies to adapt to environmental and antibiotic pressures, with specific adaptations in Bla resistance mechanisms and membrane interactions [55]. In our present work, Tet, Sul, and Bla genes have been studied, in which the long-term stability of these genes may enhance ABR because of evolutionary changes and efflux pumps. For instance, TetA and CmlA are particular to certain antibiotics and provide intrinsic resistance to bacteria via chromosomal coding [56]. Furthermore, the TetA efflux pump plays a significant role in the resistance of *E. coli* as reported by [57].

The MIC value is important for evaluating the effectiveness of antibiotics. The following antibiotics (amoxicillin, tetracycline, sulfamethoxazole, ampicillin, and chloramphenicol) were selected for monitoring resistant trends to ensure the antibiotic usage to treat bacterial (*E. coli* and *Salmonella*) infection [58]. Likewise, a study conducted in São Paulo State, Brazil, also showed that the most common resistance profile among *E. coli* isolated for tetracycline, trimethoprim–sulfadiazine, and ampicillin [59]. This indicates how important are alternative therapies such as combinational therapies, and there is a strong probability about the inefficiency of the single use of antibiotics to treat infection in the farming of tilapia production [60]. This highlights the dire need for alternative treatment strategies like

combinational therapies, as there is a possibility that the single use of antibiotics may no longer be efficient for controlling tilapia farming [61].

Synergistic combinations tend to be more effective, whereas antagonistic combinations are less potent [62]. Additive combinations are as effective in inhibiting microbial growth as the sum of individual drugs when used alone [63]. Combination therapies were evaluated in the current study to combat bacterial infections [64]. Ampicillin exerts its bactericidal influence by inhibiting the synthesis of bacterial cell walls, leading to compromised cell wall integrity and subsequent bacterial lysis [65]. Although SMX–TMP inhibits the synthesis of folic acid [66]. A combination of two antibiotics having different mechanisms of action is intended to enhance bacterial killing for the reason that when bacteria trigger to become resistant to one antibiotic, the other one is expected to restrict the bacterial infection effectively [67]. Furthermore, the inclusion of SXT–TMP in combination with either isoniazid or rifampin also effectively inhibited the development of drug resistance in vitro [66].

According to our findings, ampicillin + sulfamethoxazole showed a synergistic effect against *E. coli* and *Salmonella*. A similar study reported that sulfamethoxazole with β -lactam showed a synergistic effect against the tested strains, and no antagonistic effect was observed in β -lactam/SXT combinations [68].

Antagonism between bacteriostatic and bactericidal antibiotics is a significant factor [69] that may contribute to the failure of a particular drug combination in restricting *E. coli*. As AMR is the evolutionary mechanism that underlies the effect of a consistently changing environment [34].

Conclusion

The study reports hiked percentages of *E. coli* 32% and *Salmonella* 22% in the local fish markets. *E. coli* showed resistance against (Amoxicillin, Tetracycline, Sulfamethoxazole, Ampicillin, Chloramphenicol), and *Salmonella* showed resistance against (Amoxicillin, Tetracycline, Sulfamethoxazole, Ampicillin). The molecular and genetic study of resistant genes revealed that the Tet, Sul, and Bla gene families have a high aliphatic index, that indicates its stability. The Tet gene family exhibited remarkable conservation between the two bacterial species, specifically for TetA and TetB which showed predominant localization in the plasma membrane. Secondary structure analysis revealed that the Bla gene family has variations in residues, indicating functional divergence. For resistance modulation, we further studied antibiotic synergistic combinations for *Salmonella* (sulfa + amoxi and sulfa + ampi) and *E. coli* (sulpha + amoxi), highlighting the importance of ongoing

monitoring and interventions to combat ABR. In vitro drug combinations showed many folds of antibacterial activity with very limited antagonism. The novel drug combinations might provide insights in vivo for a wider range of treatment options for complicated cases.

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Data availability The data supporting in this study's findings are included in the manuscript and supplementary file.

Consent to Publish There are no human subjects involved in this study.

Declarations

Conflict of interest All authors declare that they have no competing interest.

Ethical Approval Not applicable.

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